PROJECT IDEA AND ABSTRACT

3-Tier Visitor & Access Management System (VAMS)

1. Project Idea

The 3-Tier Visitor & Access Management System (VAMS) is a centralized digital platform for managing who can enter an organization's premises, which areas they may access, and how long that permission remains valid. It replaces paper visitor registers and disconnected gate records with a single workflow that links user registration, manager approval, QR-based access, entry and exit logging, and live occupancy monitoring.

The system is organized around three operational tiers. The Owner (Super Admin) manages the system and its Managers. Managers review access requests and assign area permissions and validity periods. Employees, staff, guests, and temporary visitors use the system according to their account type: regular personnel sign in with their assigned credentials, while guests and temporary staff submit a registration request for review. Once an authorized Manager approves a request, VAMS generates a unique QR credential for use at the gate.

At an entry or exit point, a scanner reads the QR credential and the system validates it against the central database. A valid scan creates a timestamped event associated with the person and gate. These events are synchronized with the central data store and, if cloud hosting is implemented, the cloud service. Owner and Manager dashboards use the recorded activity to show access events and an estimated current occupancy count.

2. Problem Statement

Organizations that manage visitors manually often rely on paper registers, physical passes, phone calls, or separate gate records. These methods can make it difficult to verify approvals, apply different area restrictions, track temporary permissions, find historical entry and exit details, and determine who is currently inside the premises. In a multi-gate environment, separate or delayed records may also prevent staff from seeing a consistent view of access activity. VAMS addresses these problems by providing a centralized, role-based process for requesting, approving, issuing, validating, and recording access.

3. Project Objectives

- **Provide a single digital process** for registering employees, guests, and temporary staff and routing access requests for review.
- **Give the Owner system-wide control**, including Manager administration and access to overall activity dashboards.
- **Allow Managers to approve or reject** requests and assign Public or Restricted area access.
- **Support configurable access validity**, including One Day Only, Until Logged Off, and Ongoing, according to the implemented rules.
- **Generate a unique QR credential** after approval and validate it at authorized gate scanning points.
- **Record entry and exit events** with person, gate, scan type, and timestamp details.
- **Present access activity and calculated occupancy information** on Owner and Manager dashboards.

4. System Overview

Tier 1: Owner / Super Admin

The Owner is the highest-privilege user. This tier creates or assigns Manager accounts, manages or resets Manager credentials through authorized procedures, and reviews overall system activity. The Owner dashboard provides a system-wide view of access requests, users, scan activity, and the occupancy value calculated from recorded entry and exit events.

Tier 2: Manager / Admin

Managers handle day-to-day access administration. They review pending requests, confirm the requested purpose and user details, and decide whether access should be granted. For approved requests, Managers assign an area category—Public or Restricted—and a validity option such as One Day Only, Until Logged Off, or Ongoing. Their permissions are limited to the actions granted by the system's role policy.

Tier 3: Employees, Guests, and Temporary Staff

Regular employees and staff use their assigned ID and password to access the system. Guests and temporary personnel, such as visitors, mechanics, contractors, or delivery personnel, submit a registration or access-request form. Their access remains pending until an authorized Manager reviews it. Approved users receive a QR credential for presentation at the gate.

5. Main System Workflow

1. **Registration and request creation:** A staff member signs in with assigned credentials, while a guest or temporary visitor submits a registration form. The system stores the relevant user details and creates an access request for approval.
2. **Manager review and decision:** A Manager views pending requests, reviews the submitted details, and approves or rejects the request. For approval, the Manager sets the area permission and validity period.
3. **QR credential issuance:** After approval, the system generates a unique QR code and associated identifier. The credential is made available to the approved user through the configured interface.
4. **Gate verification:** At an entry or exit point, a scanner reads the QR code and submits its token for server-side validation. An accepted scan is recorded with the gate and timestamp; an invalid, unapproved, or expired credential is denied.
5. **Monitoring and occupancy:** Scan events are stored centrally and made available to management dashboards. The system calculates occupancy from recorded entry and exit activity and presents the result to Owner and Manager users.

6. Abstract

The 3-Tier Visitor & Access Management System (VAMS) is proposed as a centralized solution for controlling and monitoring access to organizational premises. The system separates responsibilities among an Owner (Super Admin), Managers, and end users such as employees, guests, and temporary staff. Regular personnel authenticate using assigned credentials, while visitors submit an access request for review. Managers evaluate pending requests and, when granting access, assign an area permission and a validity period. After approval, the system generates a unique QR credential for presentation at designated gates. The scanner validates the credential and records entry or exit activity with the relevant user, gate, and timestamp. Centralized records support Owner and Manager dashboards that display access activity and an occupancy figure derived from scan events. By combining role-based administration, configurable access permissions, QR-based verification, and timestamped records, VAMS aims to improve the organization and traceability of visitor access compared with manual register-based processes. The system's occupancy figure depends on accurate and complete scans, and its effectiveness must be assessed through implementation and testing.

7. Scope of the Project

The proposed scope includes role-based accounts for Owners, Managers, employees, guests, and temporary staff; visitor registration and request review; approval or rejection; Public and Restricted area categorization; validity selection; QR credential generation and scanning; entry and exit timestamp recording; a central data store; and Owner and Manager dashboards with occupancy monitoring.

The following capabilities should be included only if they are implemented and verified in the actual project: cloud synchronization, physical turnstile or door-lock control, biometric identification, identity-document verification, native mobile applications, SMS/push notifications, and live GPS tracking. A QR-based prototype can record and report access events without directly operating a physical lock.

8. Assumptions and Clarifications

- The project description mentions both a central database and cloud synchronization. The final report should state whether the implemented database is local, hosted, or synchronized with a separate cloud service.
- The phrase “Until Logged Off” needs a defined rule: specify whether this refers to a user account session, a visit session, or another event, and ensure the scanner enforces that rule.
- Entry and exit scans must be distinguishable. If the project uses separate entry/exit gates, the gate configuration should identify direction; if it toggles direction from the previous scan, missed or concurrent scans may affect accuracy.
- Occupancy is an estimate based on recorded scans. It is accurate only when scans are complete and entry/exit events are correctly classified.
- Manager password resets and other privileged account changes should follow an authorized process and should not expose passwords in plain text.

9. Keywords

Visitor management, access control, role-based access, QR code, access request, gate scanning, entry and exit timestamp, occupancy monitoring, centralized database.

10. Architecture Reference

This text version represents the supplied system description for use where the original image is unreadable. It is a written interpretation of the user-provided project information, not a verified diagram of implemented software.

- **Owner / Super Admin** → manages Managers and monitors system activity
- **Employees and staff** → sign in with assigned ID and password
- **Guests and temporary staff** → submit registration/access request
- **Registration and request data** → Central Database
- **Manager** → reviews pending request → assigns Public/Restricted area and validity → approves or rejects
- **Approved request** → unique QR code and associated ID generated → credential delivered to user
- **User presents QR** → Entry/Exit gate scanner validates credential → timestamped scan event stored centrally
- **Central scan records** / cloud service, if implemented → occupancy calculation and activity metrics → Owner and Manager dashboards

11. References and Source Basis

Primary source: the project concept and written description supplied by the user under the title "3-Tier Visitor & Access Management System (VAMS)." The project idea, abstract, tier descriptions, workflow, scope, assumptions, and text-based architecture above are derived from that supplied material. The original image was not used as a factual source because the user reported that its text was not readable.

If the final submission requires external references, add the actual technology documentation used by the implementation (for example, Java/Spring, database, QR-code library, and browser-scanning library documentation). Do not list technologies as implemented unless they are present in the actual project.